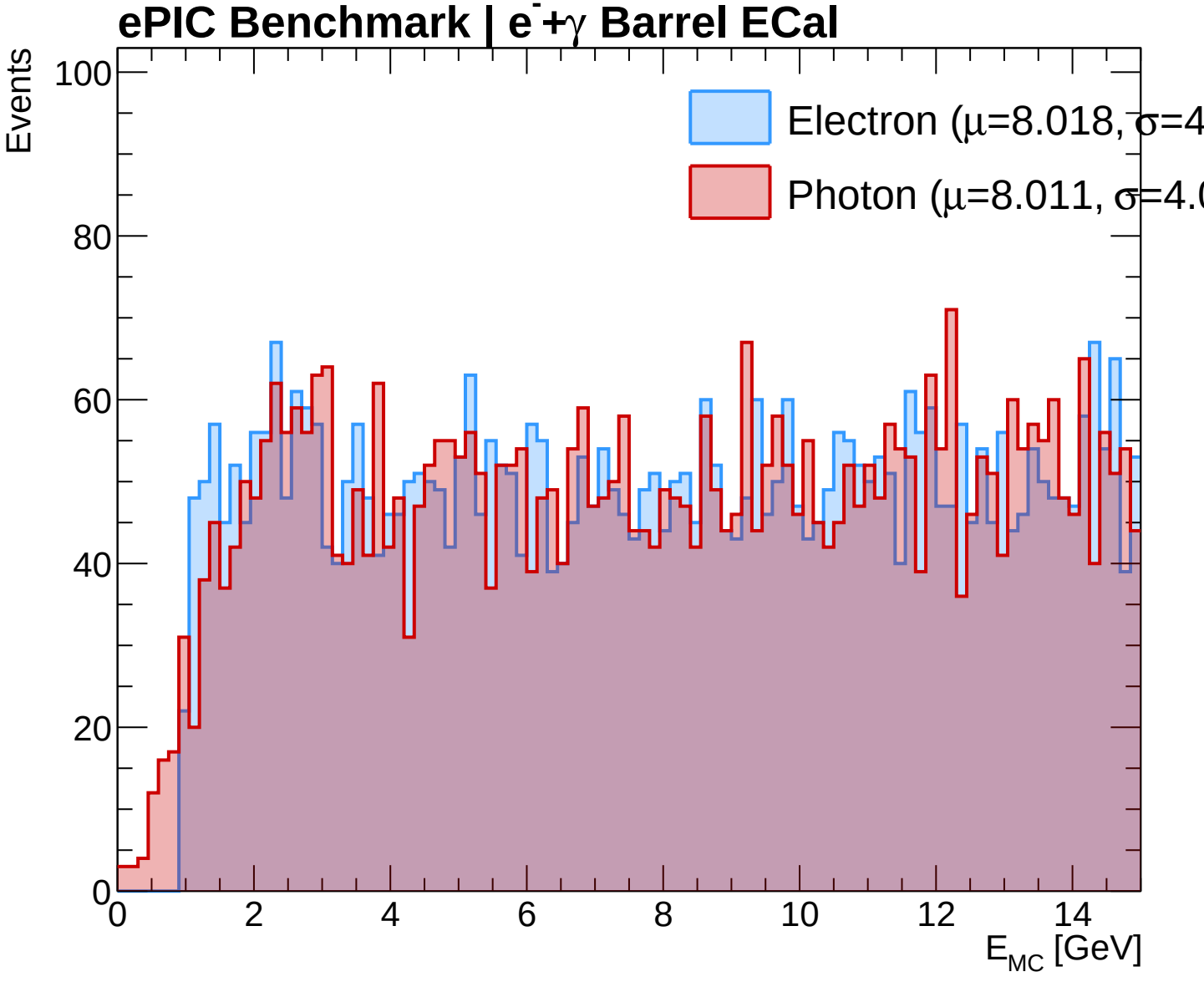
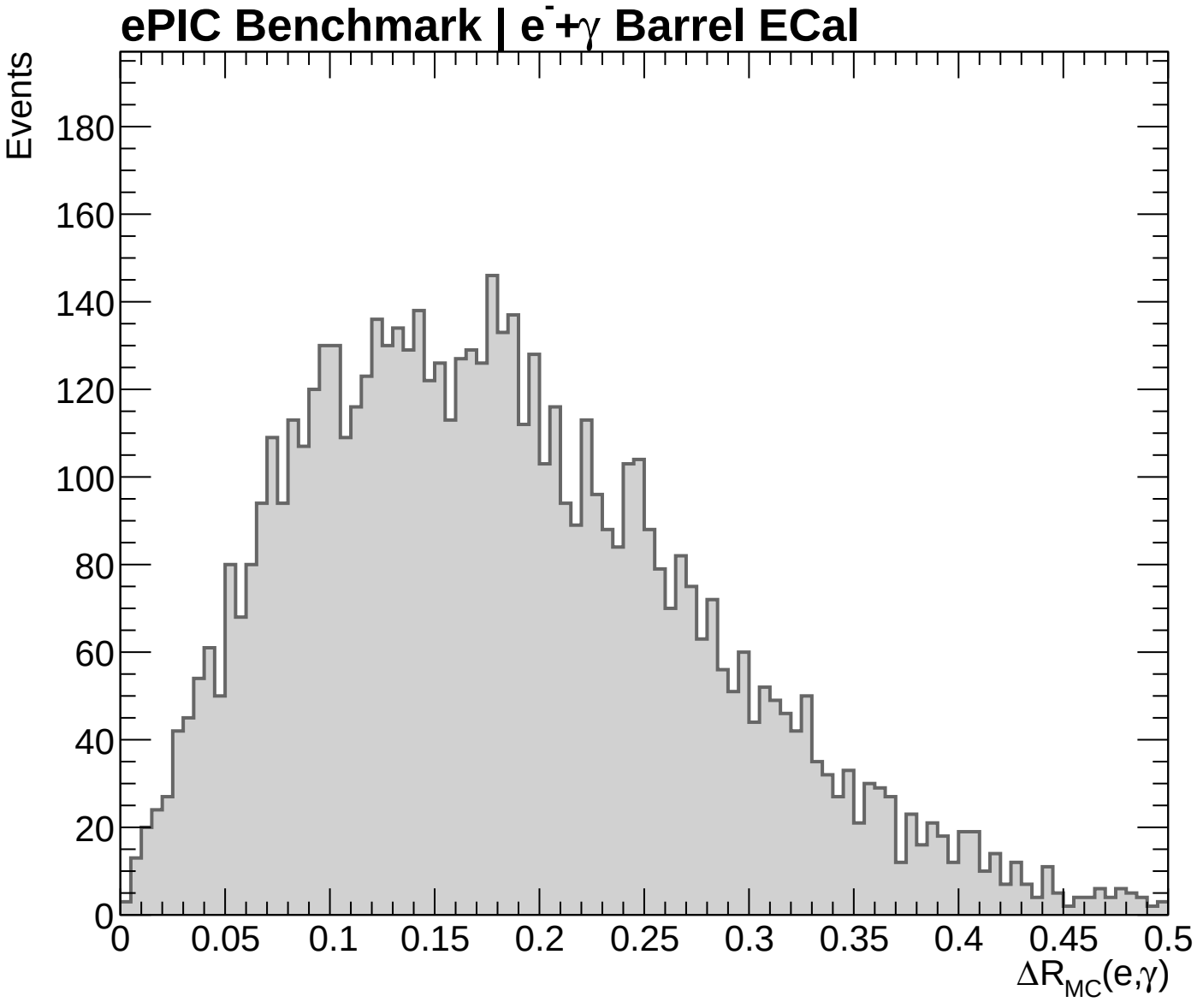
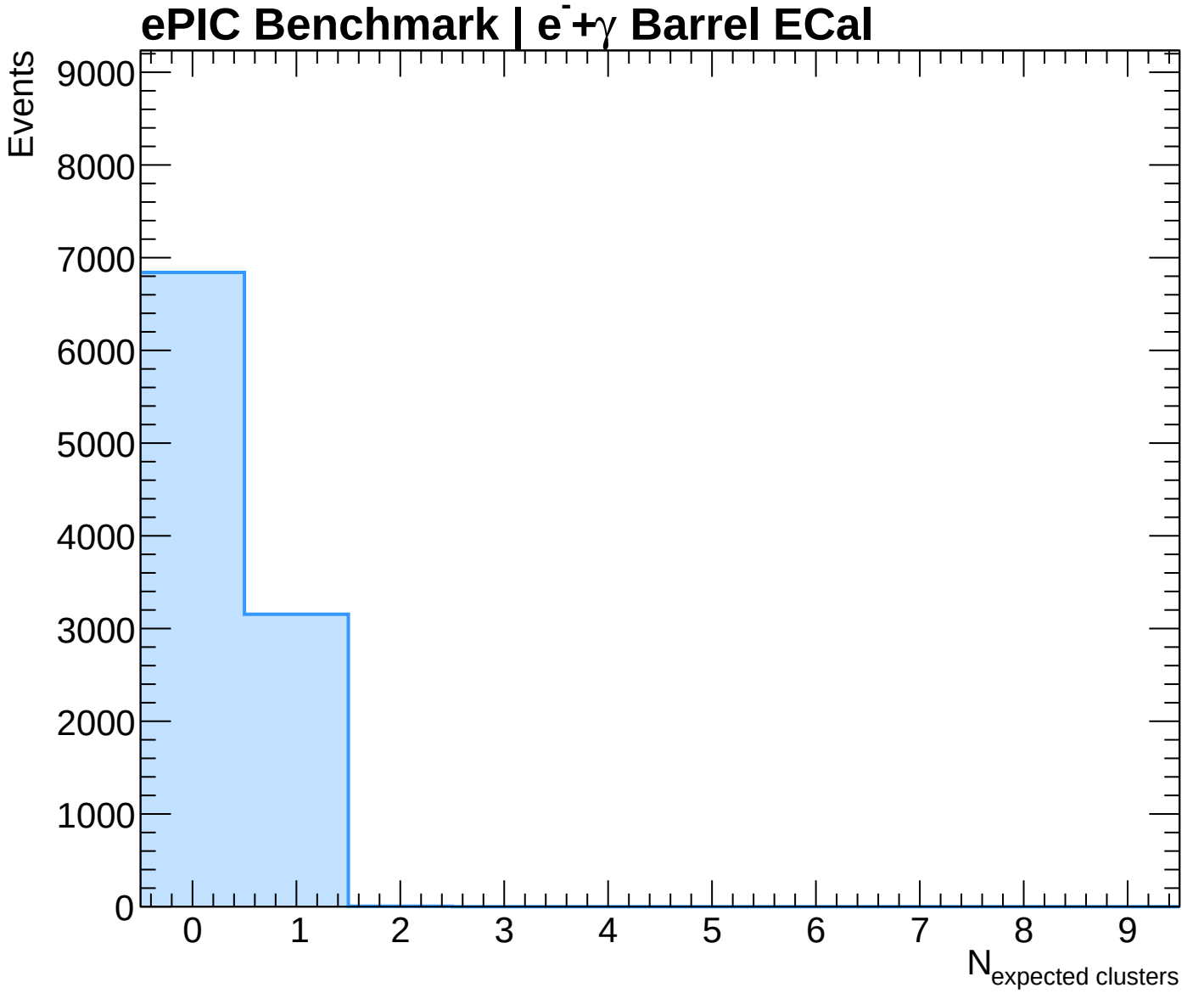
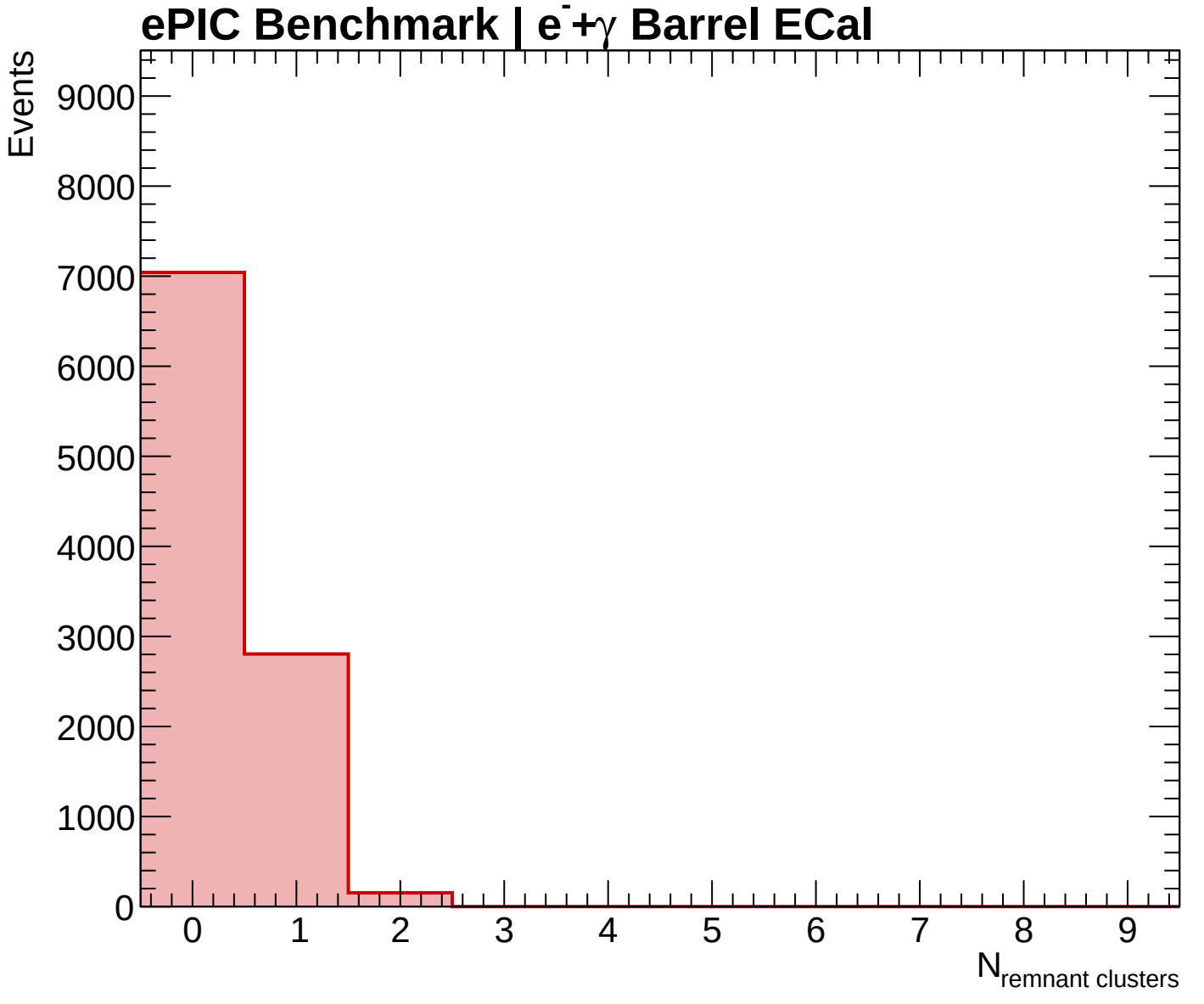
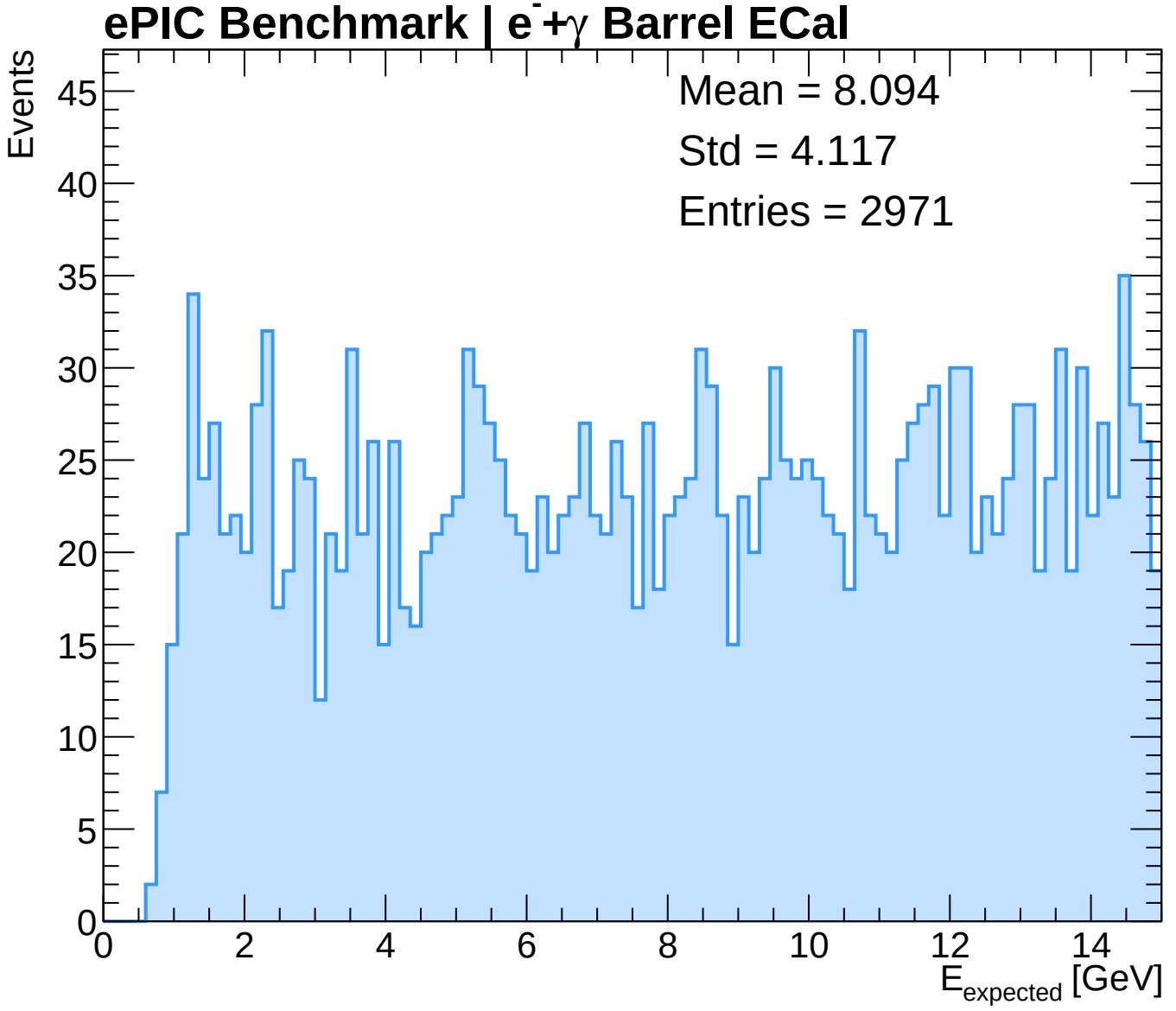
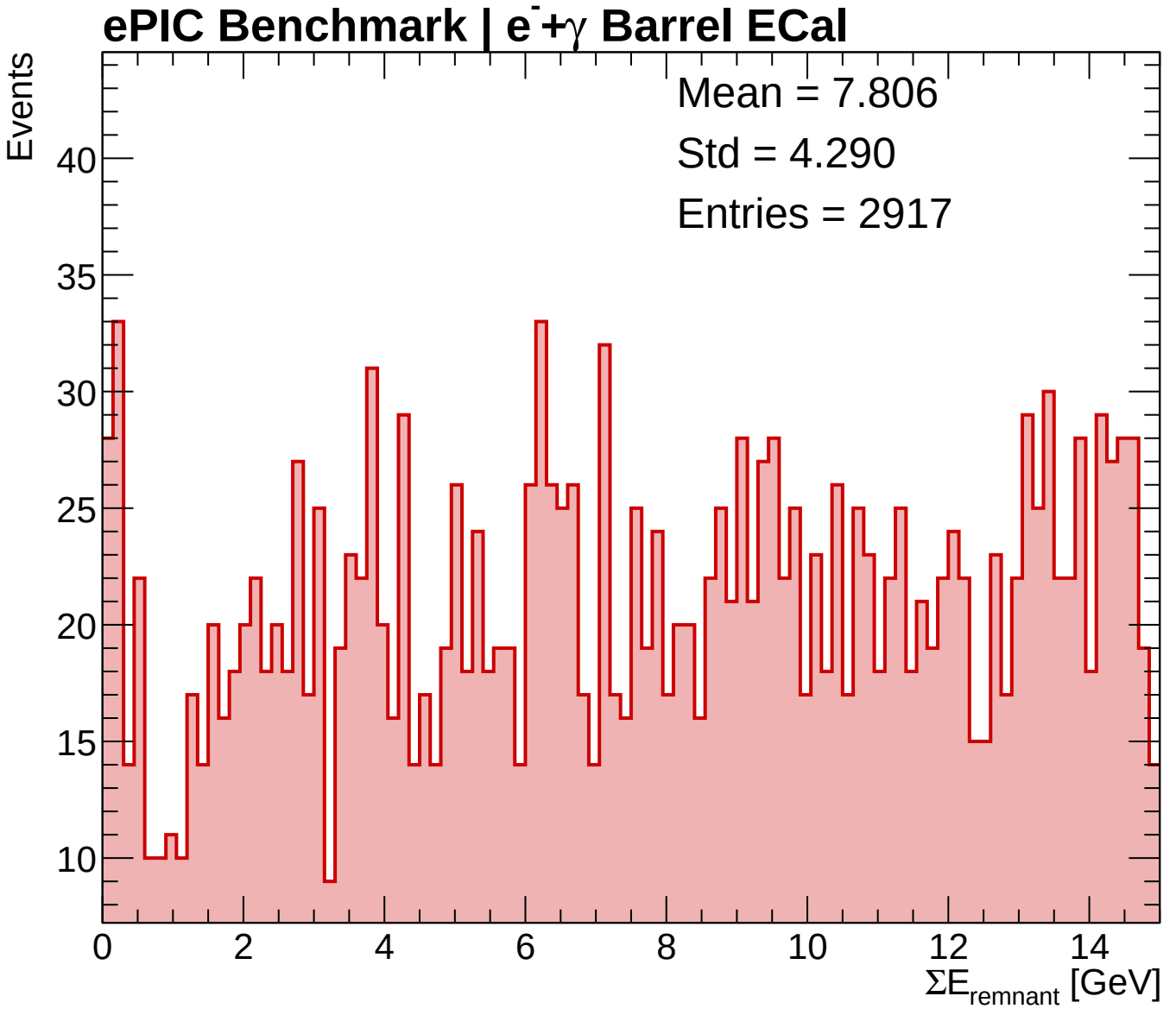

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Electron (μ=8.018, σ=4
Photon (μ=8.011, σ=4.0
0
20
40
60
80
100
0
2
4
6
8
10
12
14
E_MC [GeV]

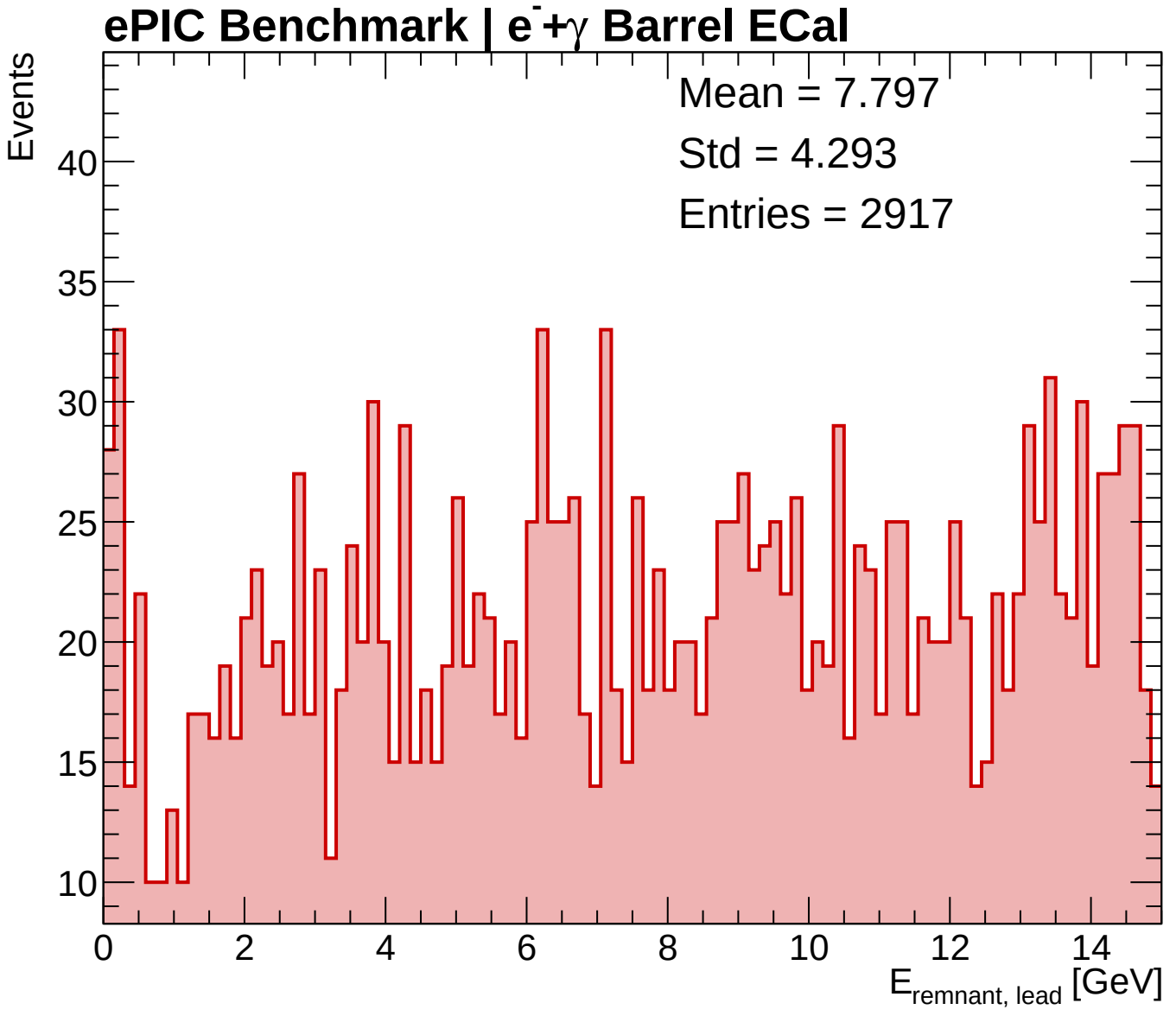

ePIC Benchmark | e⁻+γ Barrel ECal
Events
0
20
40
60
80
100
120
140
160
180
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)

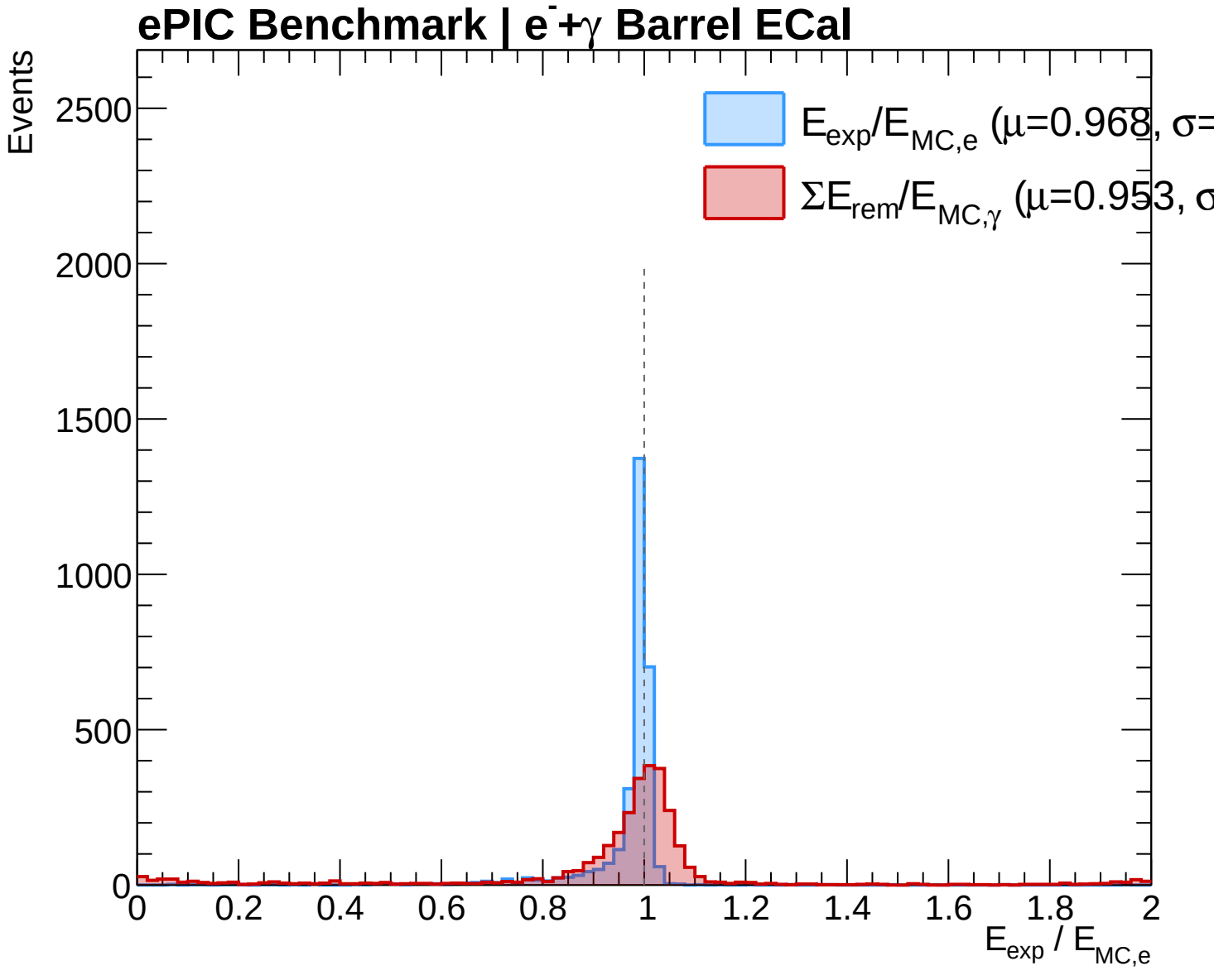

ePIC Benchmark | e⁻+γ Barrel ECal
Events
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
N expected clusters

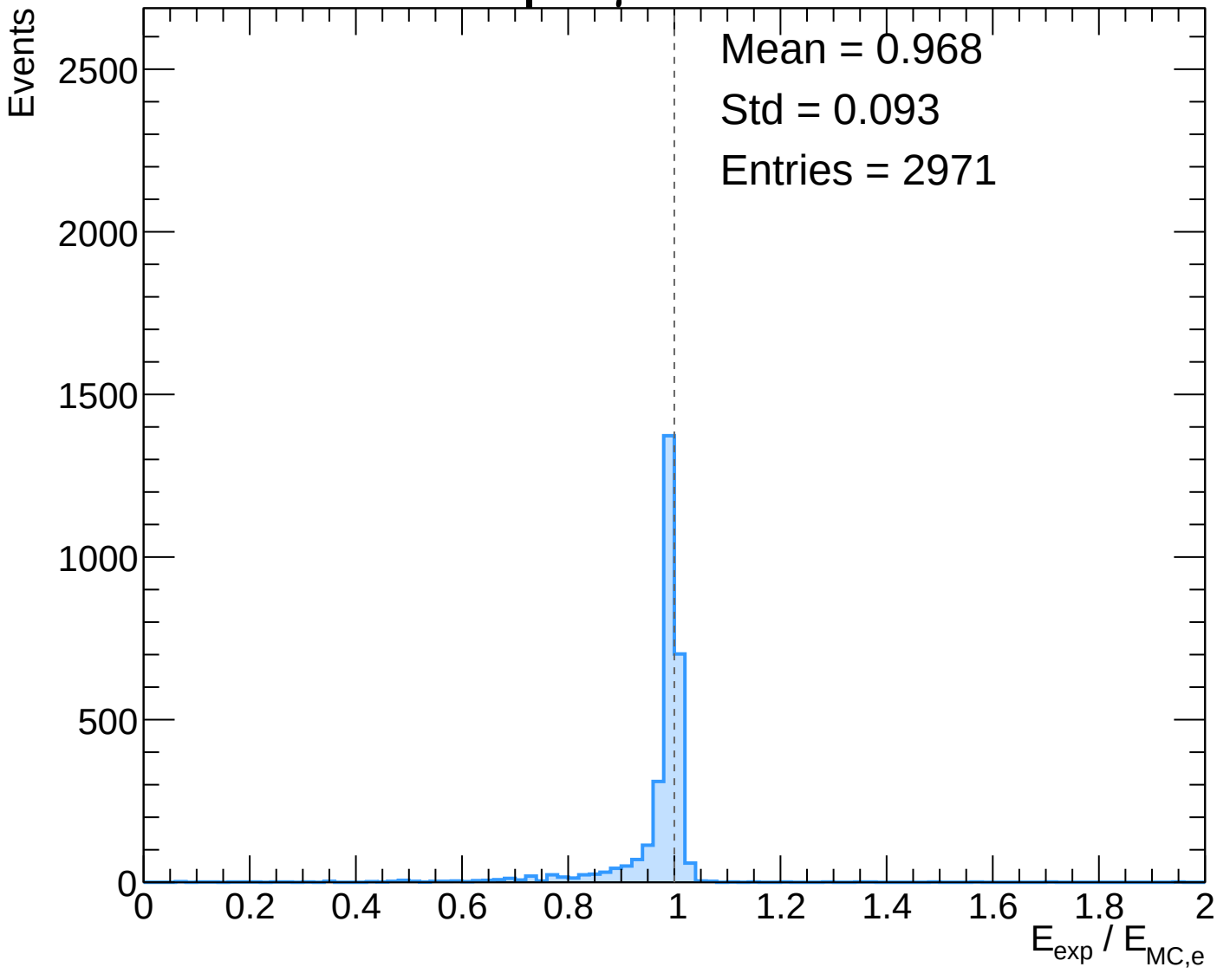

ePIC Benchmark | e⁻+γ Barrel ECal
Events
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
N remnant clusters

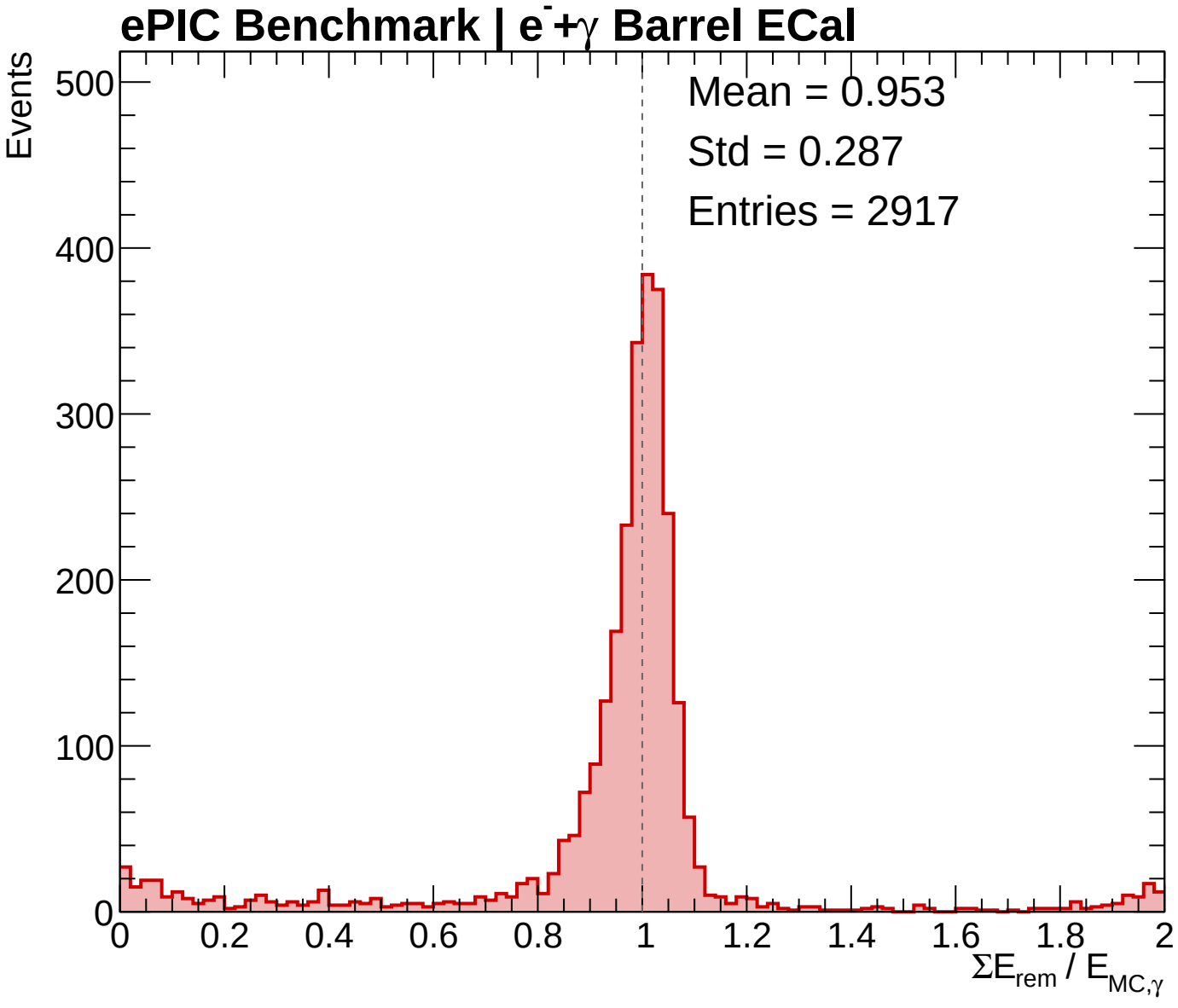

ePIC Benchmark | e⁻+γ Barrel ECal
Mean = 8.094
Std = 4.117
Entries = 2971
Events
0
5
10
15
20
25
30
35
40
45
0
2
4
6
8
10
12
14
E_expected [GeV]

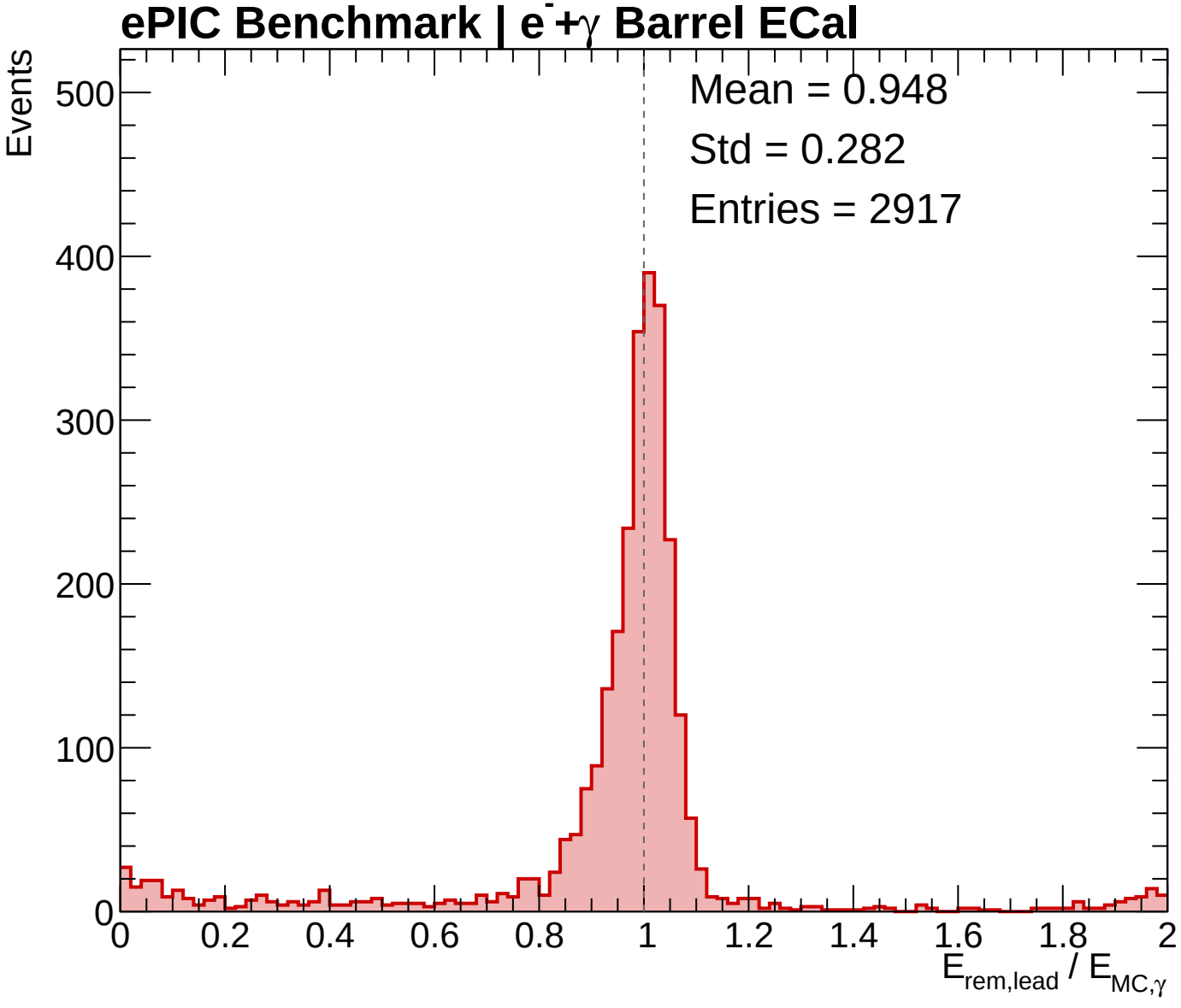

ePIC Benchmark | e⁻+γ Barrel ECal
Mean = 7.806
Std = 4.290
Entries = 2917
Events
40
35
30
25
20
15
10
0
2
4
6
8
10
12
14
ΣE_remnant [GeV]

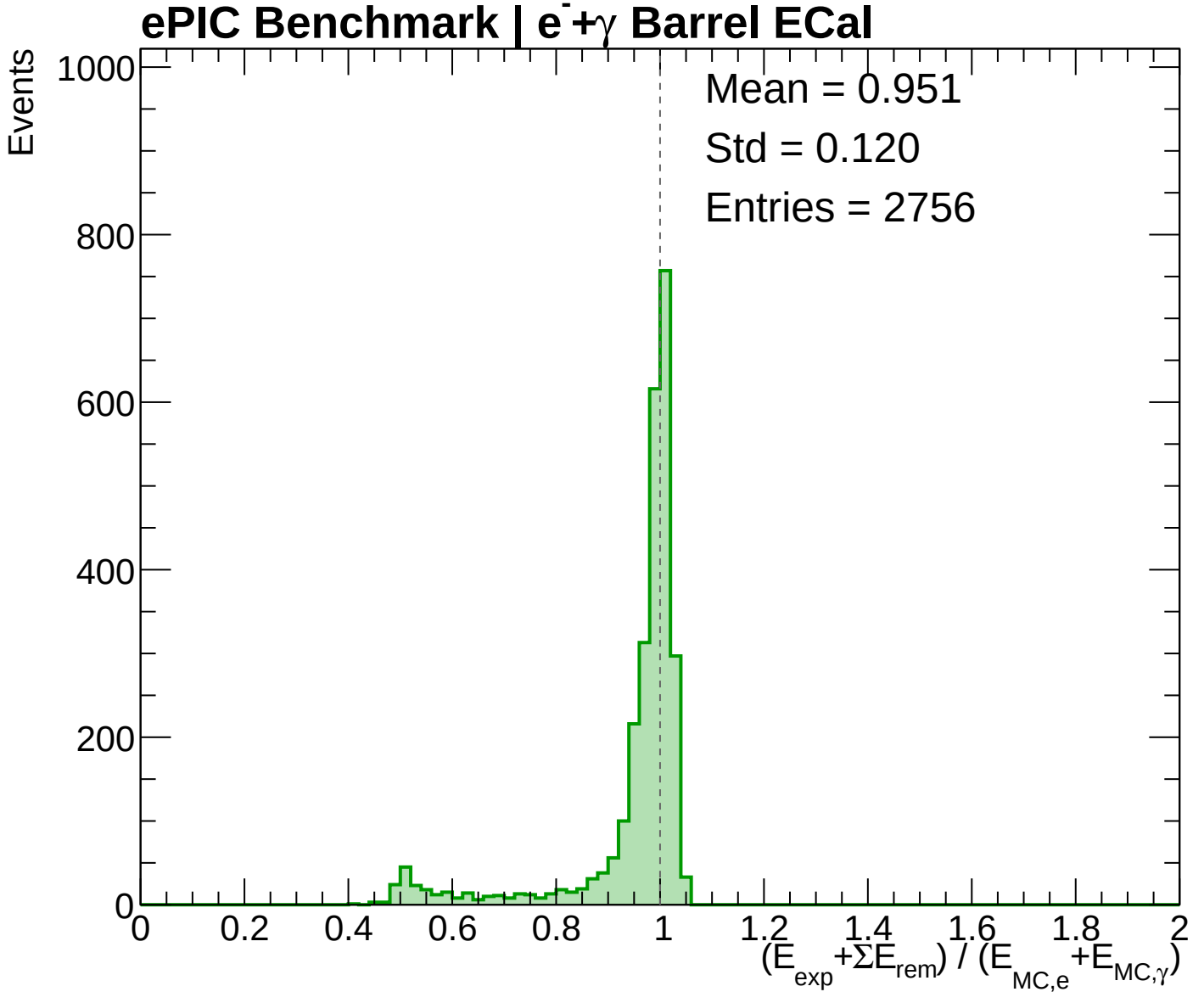

ePIC Benchmark | e⁻+γ Barrel ECal
Mean = 7.797
Std = 4.293
Entries = 2917
Events
40
35
30
25
20
15
10
0
2
4
6
8
10
12
14
E remnant, lead [GeV]

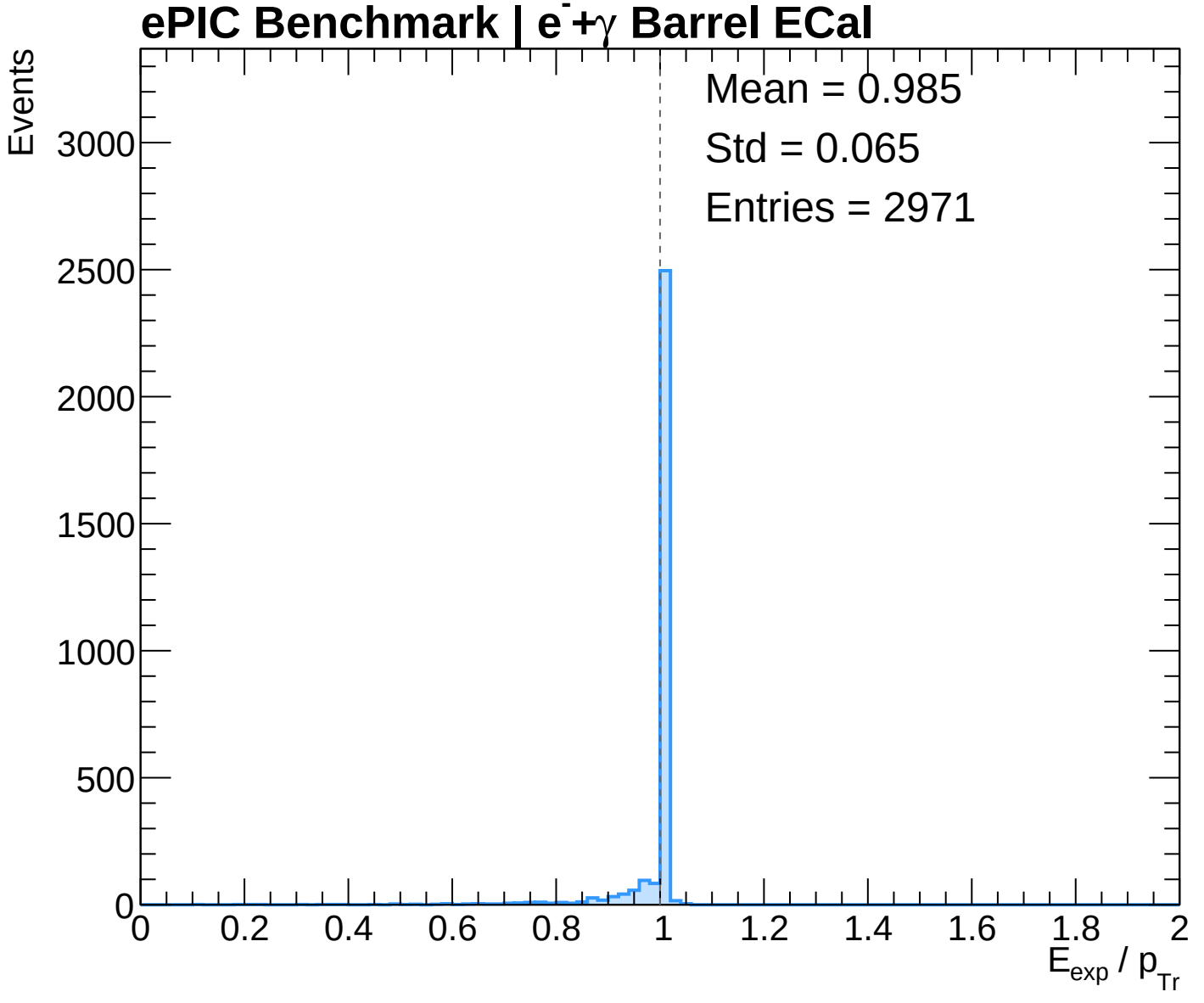

ePIC Benchmark | e⁻+γ Barrel ECal
Events
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / E_MC,e
E_exp/E_MC,e (μ=0.968, σ=
ΣE_rem/E_MC,γ (μ=0.953, σ

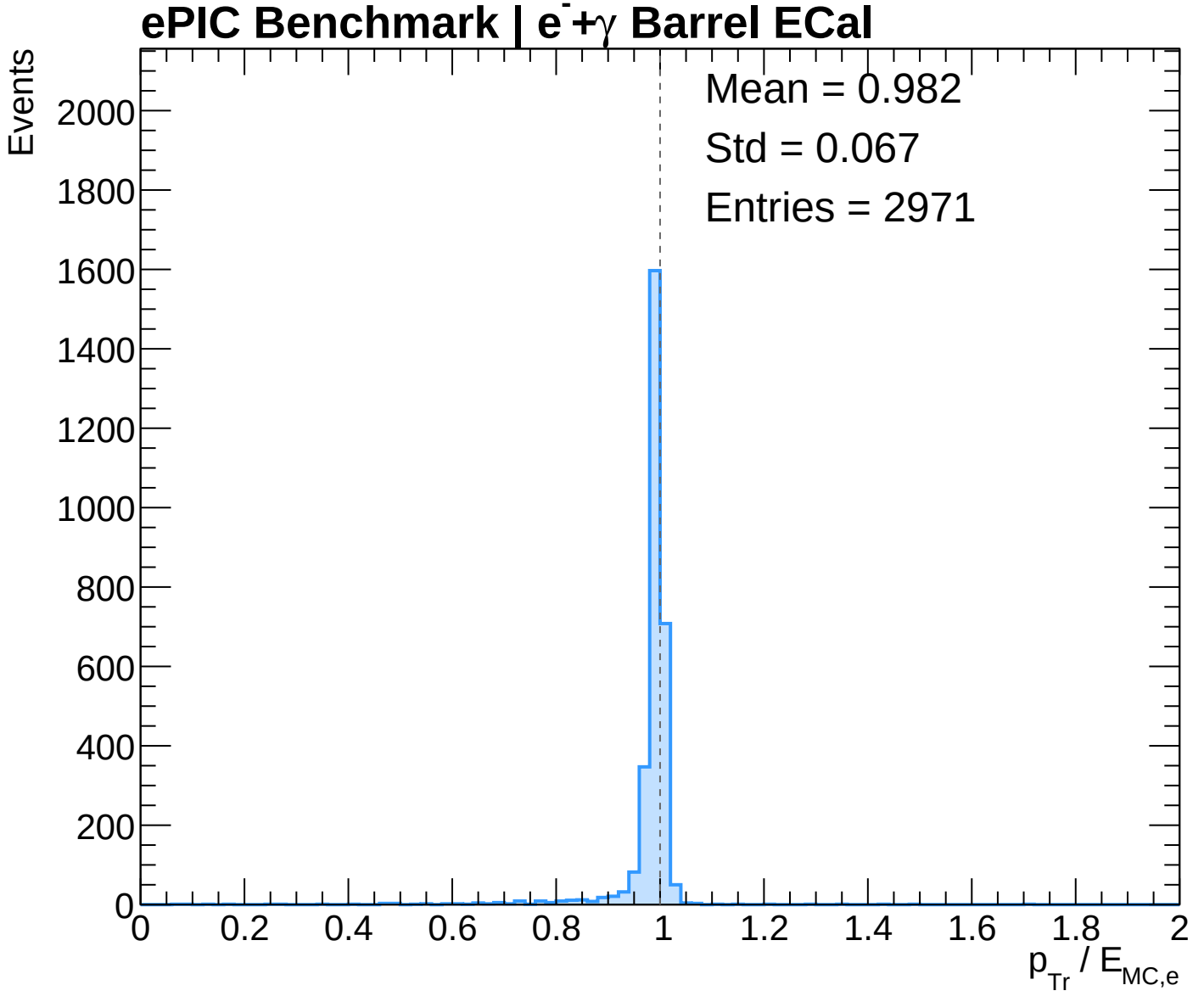

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Mean = 0.968
Std = 0.093
Entries = 2971
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / E_MC,e

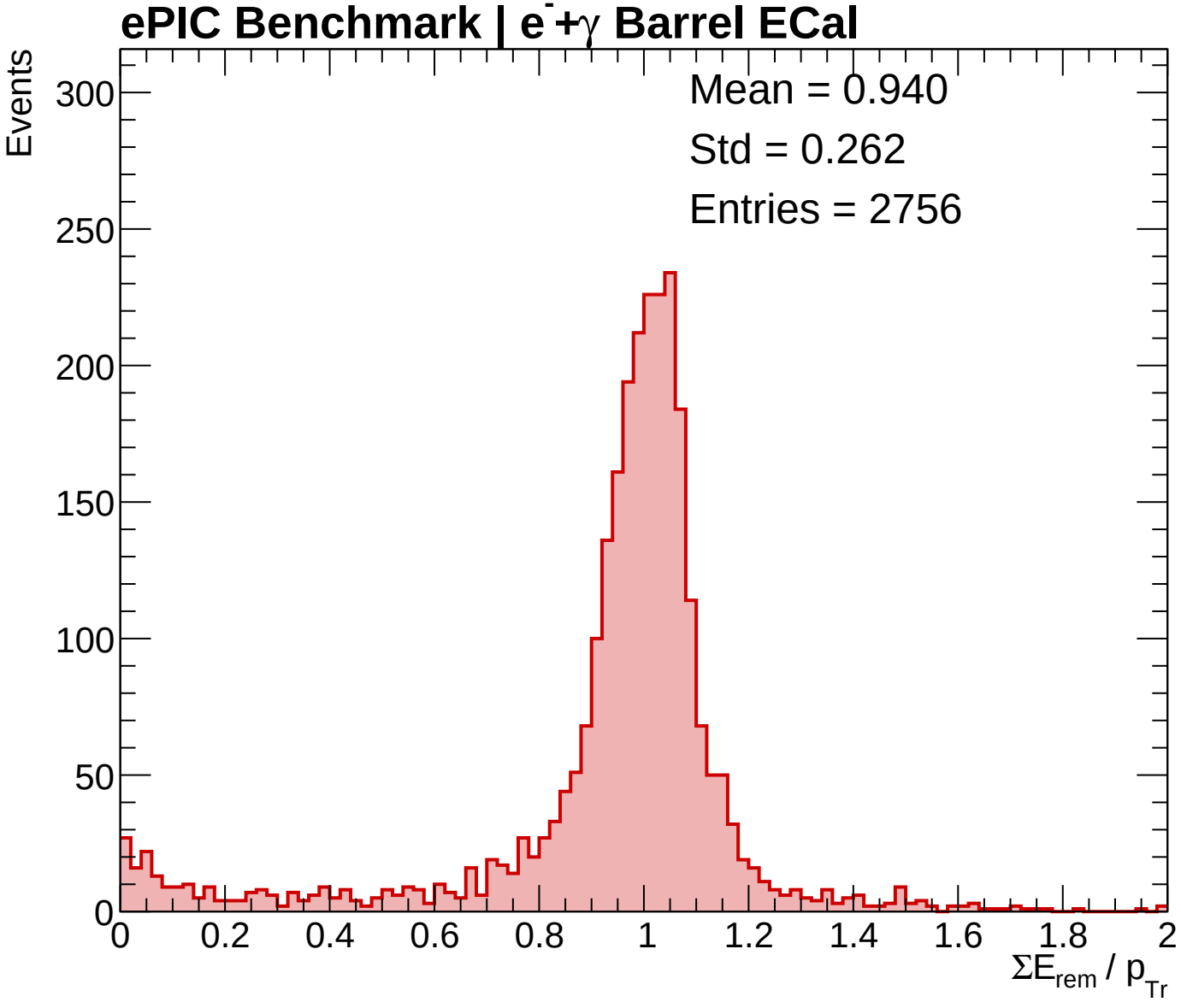

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Mean = 0.953
Std = 0.287
Entries = 2917
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΣE_rem / E_MC,γ

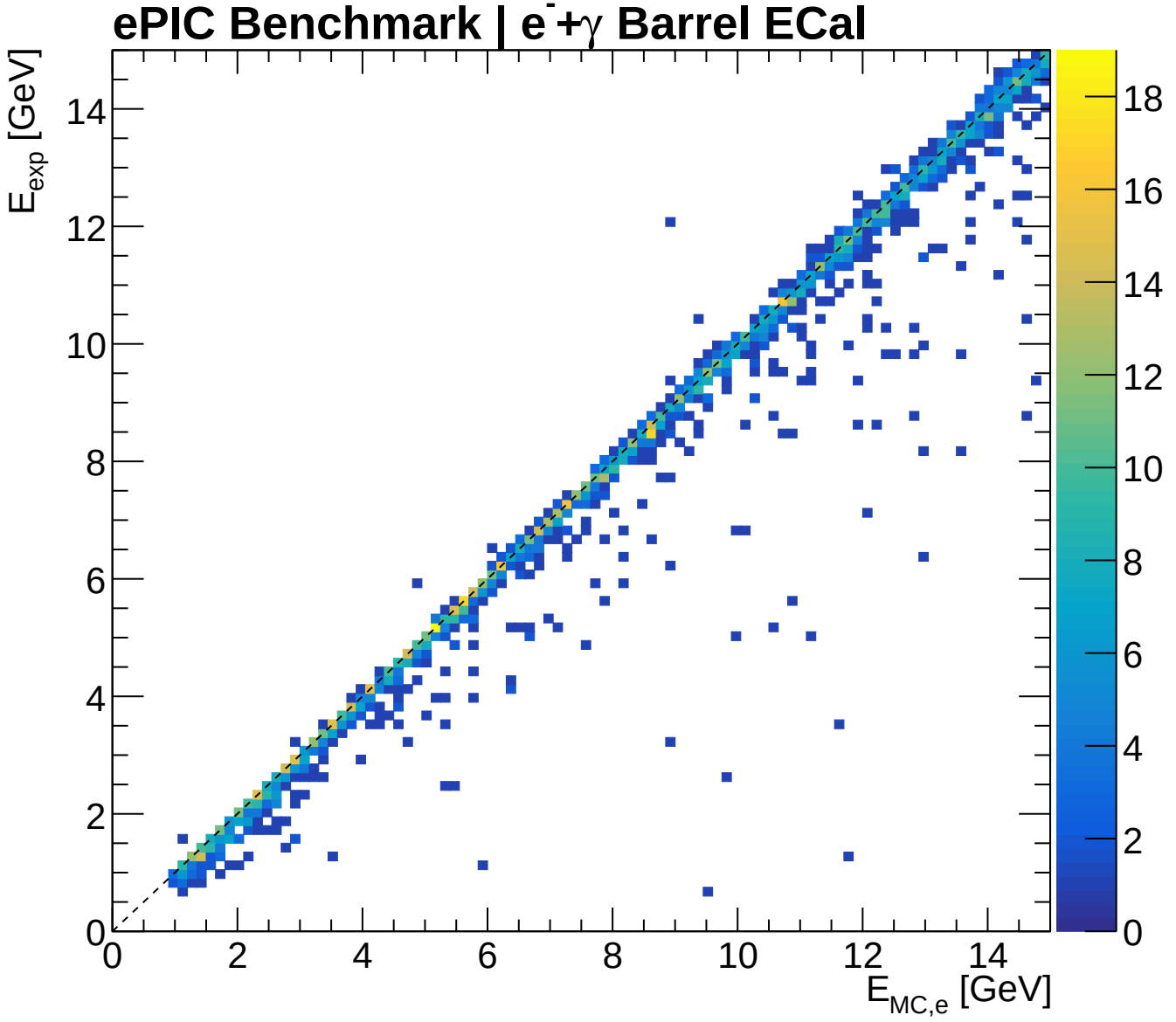

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Mean = 0.948
Std = 0.282
Entries = 2917
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_rem,lead / E_MC,γ

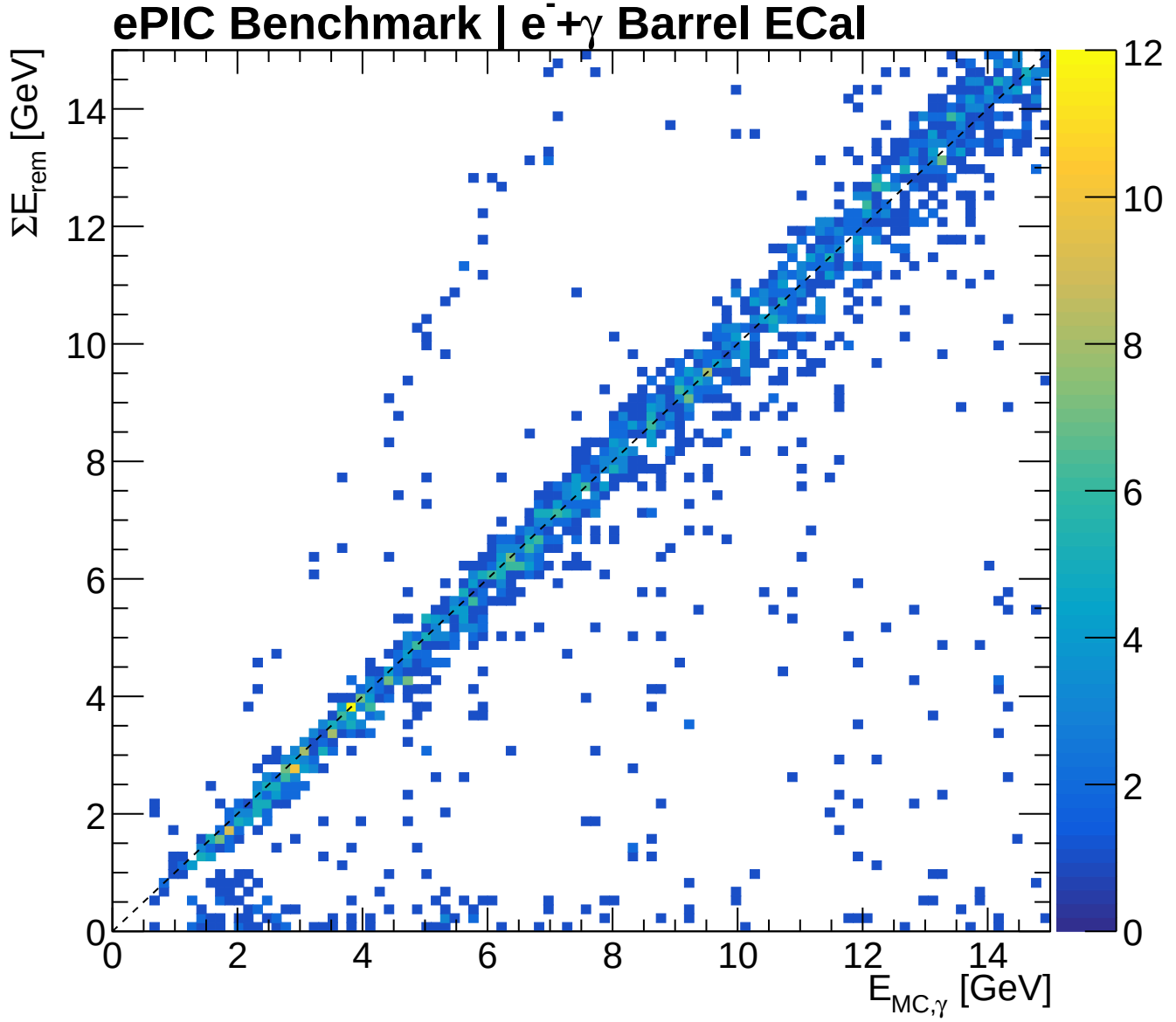

ePIC Benchmark | e⁻+γ Barrel ECal
Events
1000
800
600
400
200
0
Mean = 0.951
Std = 0.120
Entries = 2756
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+ΣE_rem) / (E_MC,e+E_MC,γ)

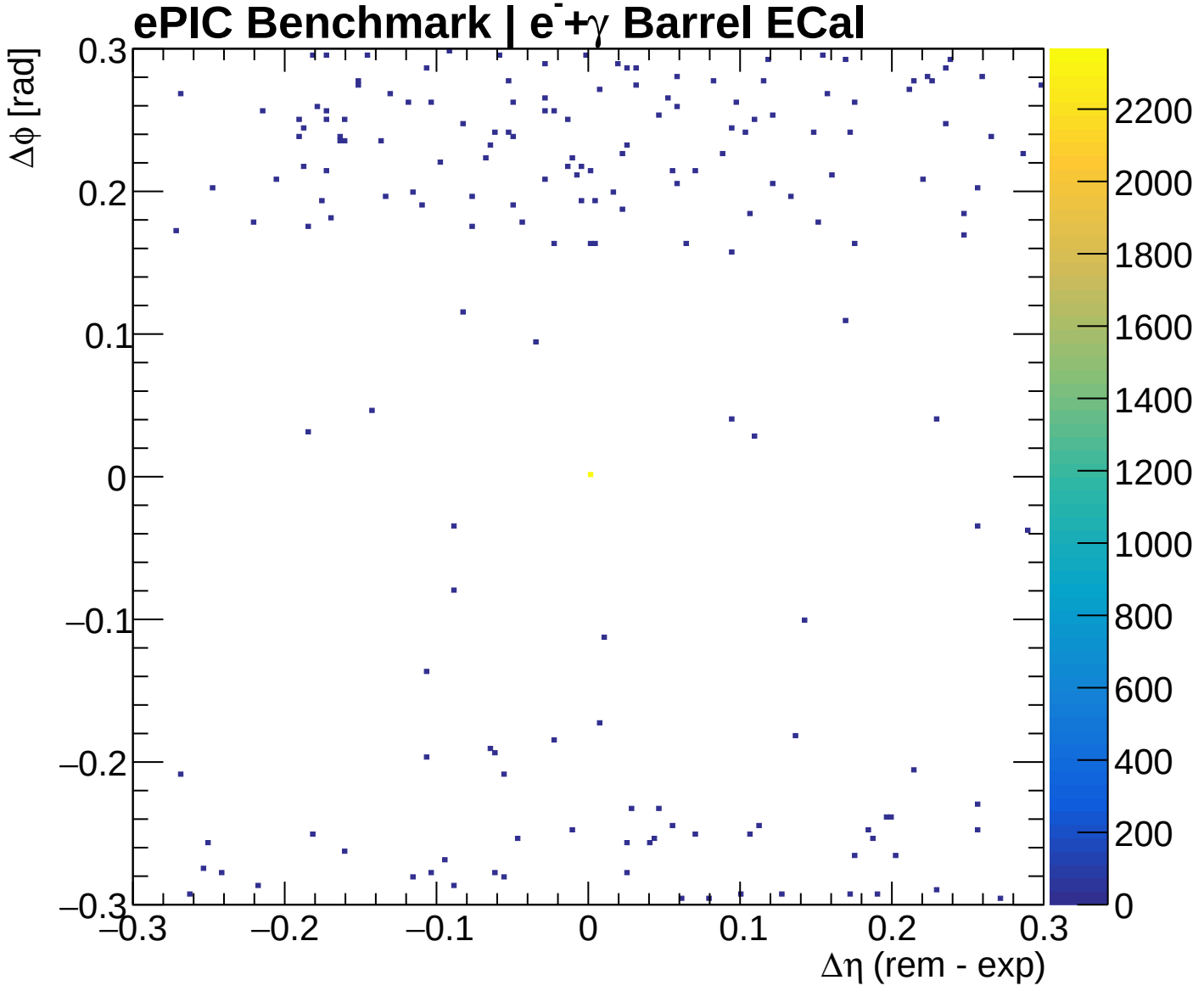

ePIC Benchmark | e⁻+γ Barrel ECal
Mean = 0.985
Std = 0.065
Entries = 2971
Events
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

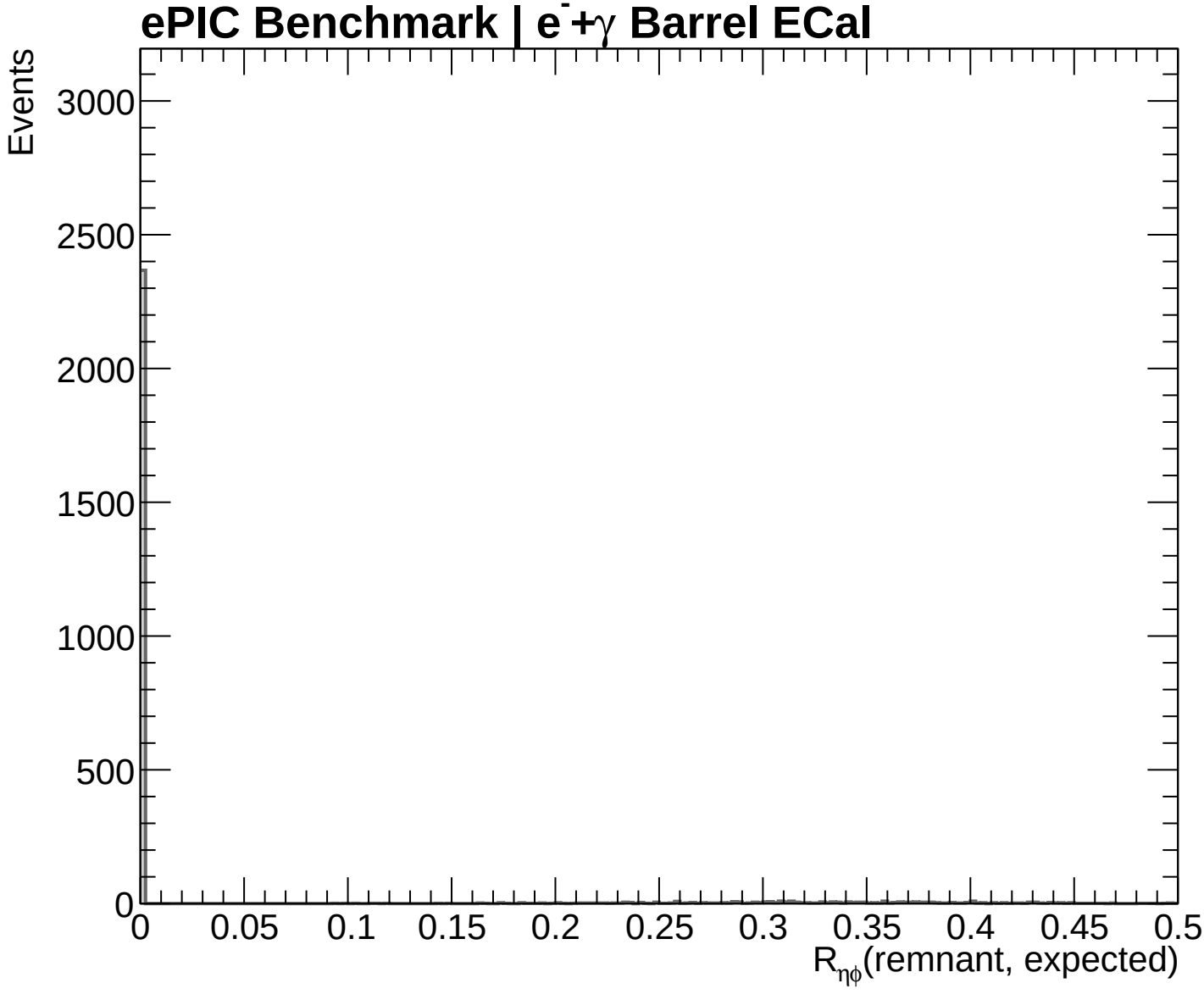

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Mean = 0.982
Std = 0.067
Entries = 2971
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
p_Tr / E_MC,e

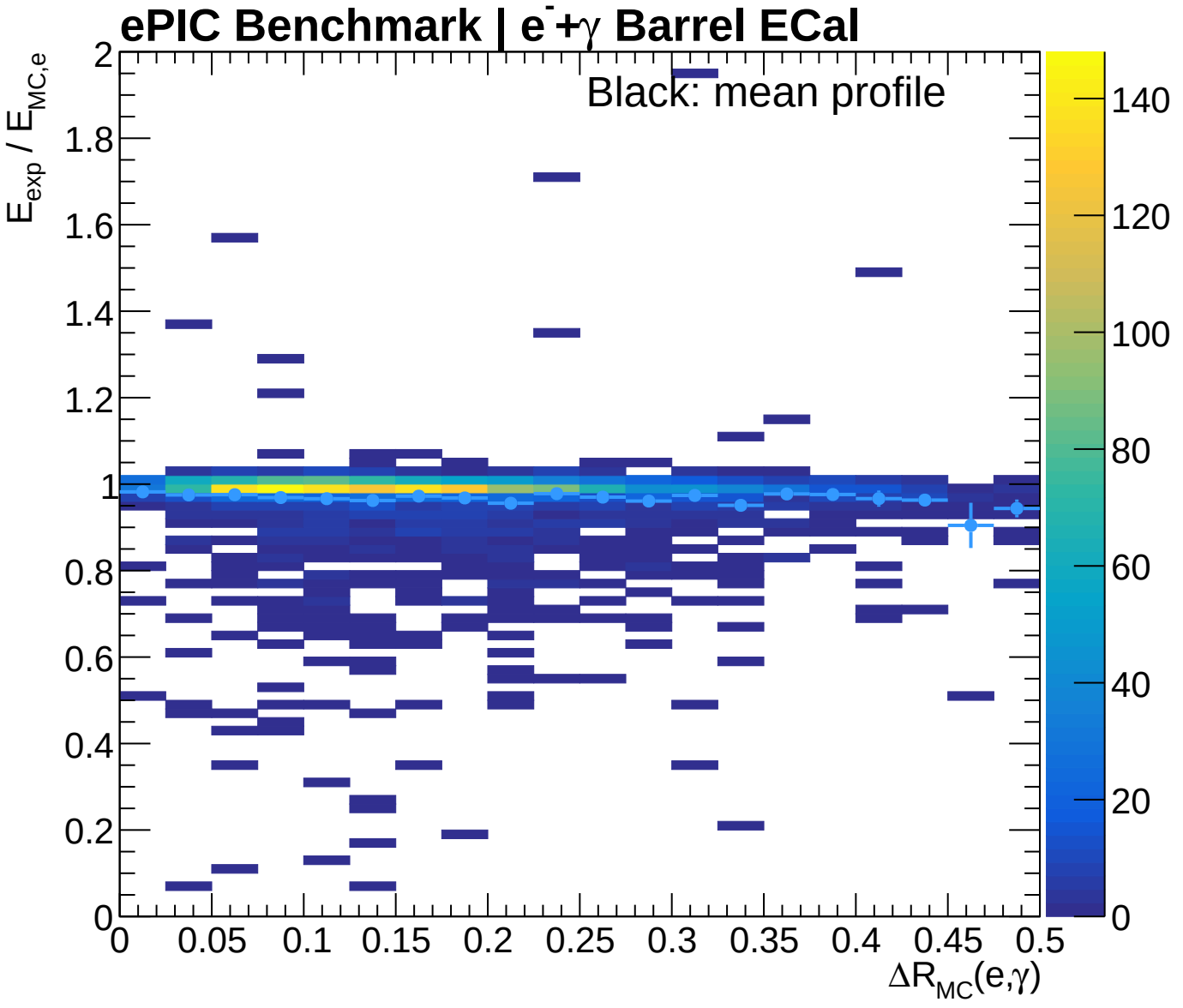

ePIC Benchmark | e⁻+γ Barrel ECal
Events
Mean = 0.940
Std = 0.262
Entries = 2756
300
250
200
150
100
50
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΣE_rem / p_Tr

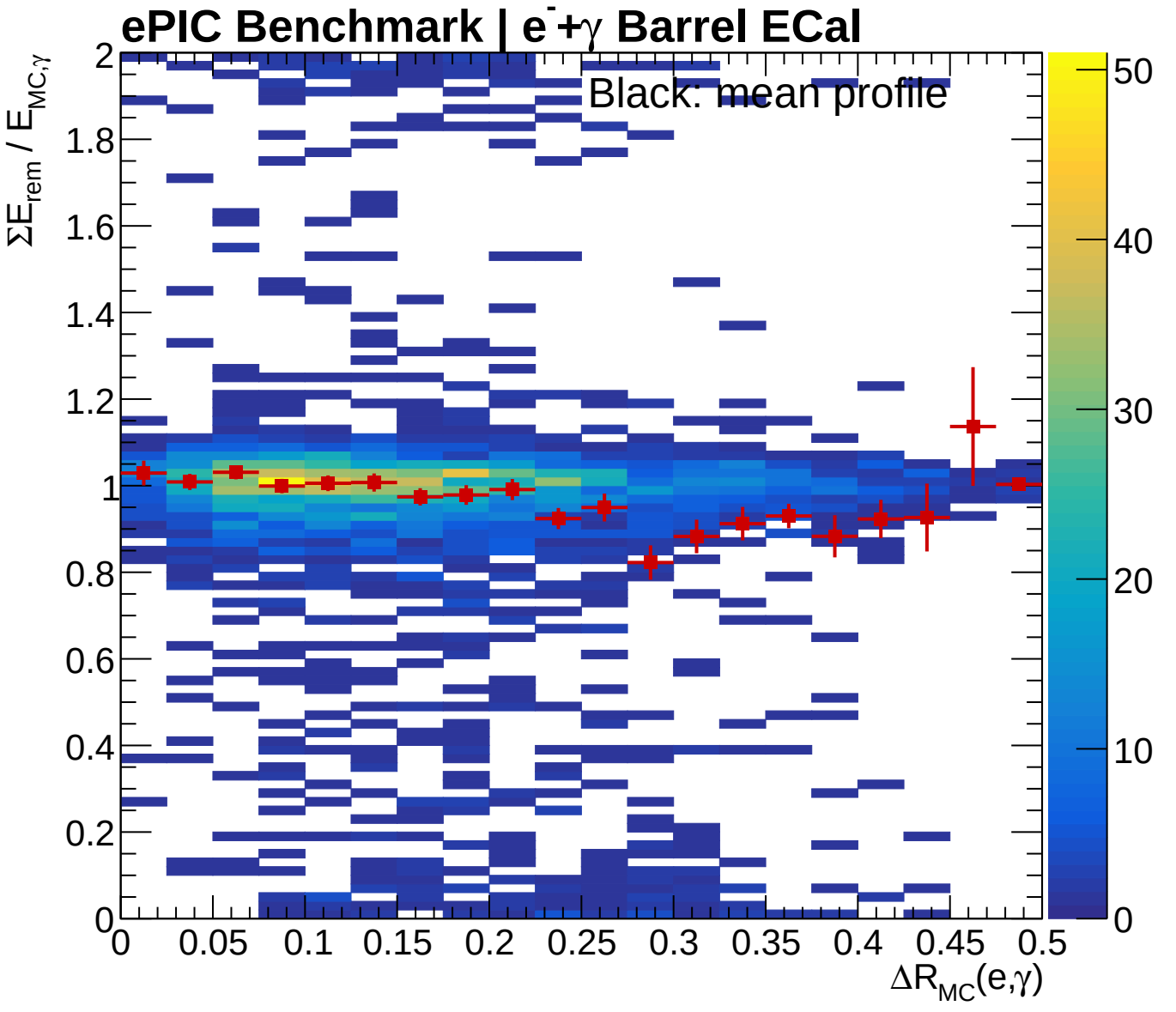

ePIC Benchmark | e⁻+γ Barrel ECal
E_exp [GeV]
E_MC,e [GeV]
0
2
4
6
8
10
12
14
16
18

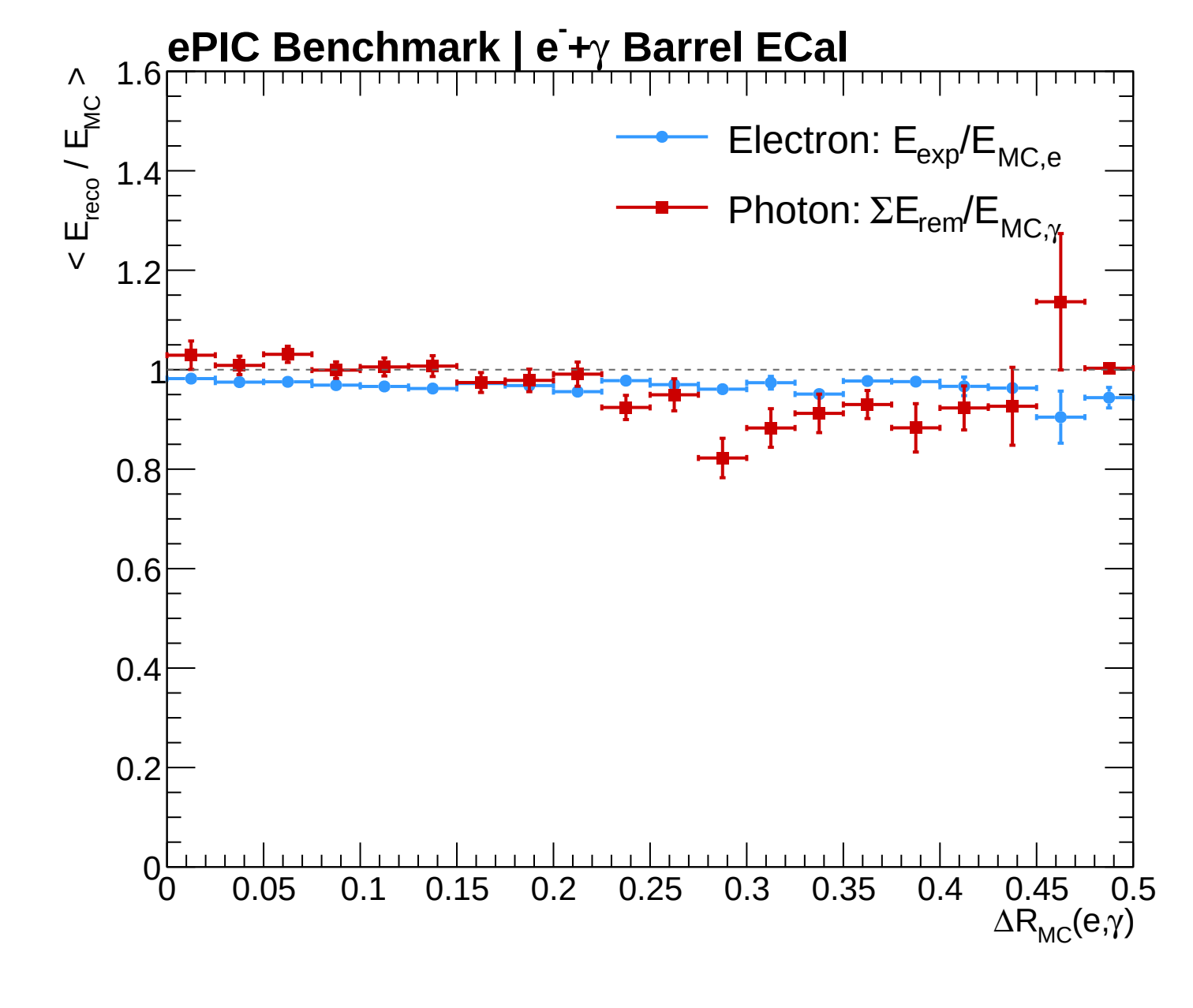

ePIC Benchmark | e⁻+γ Barrel ECal
ΣE_rem [GeV]
E_MC,γ [GeV]
0
2
4
6
8
10
12
14

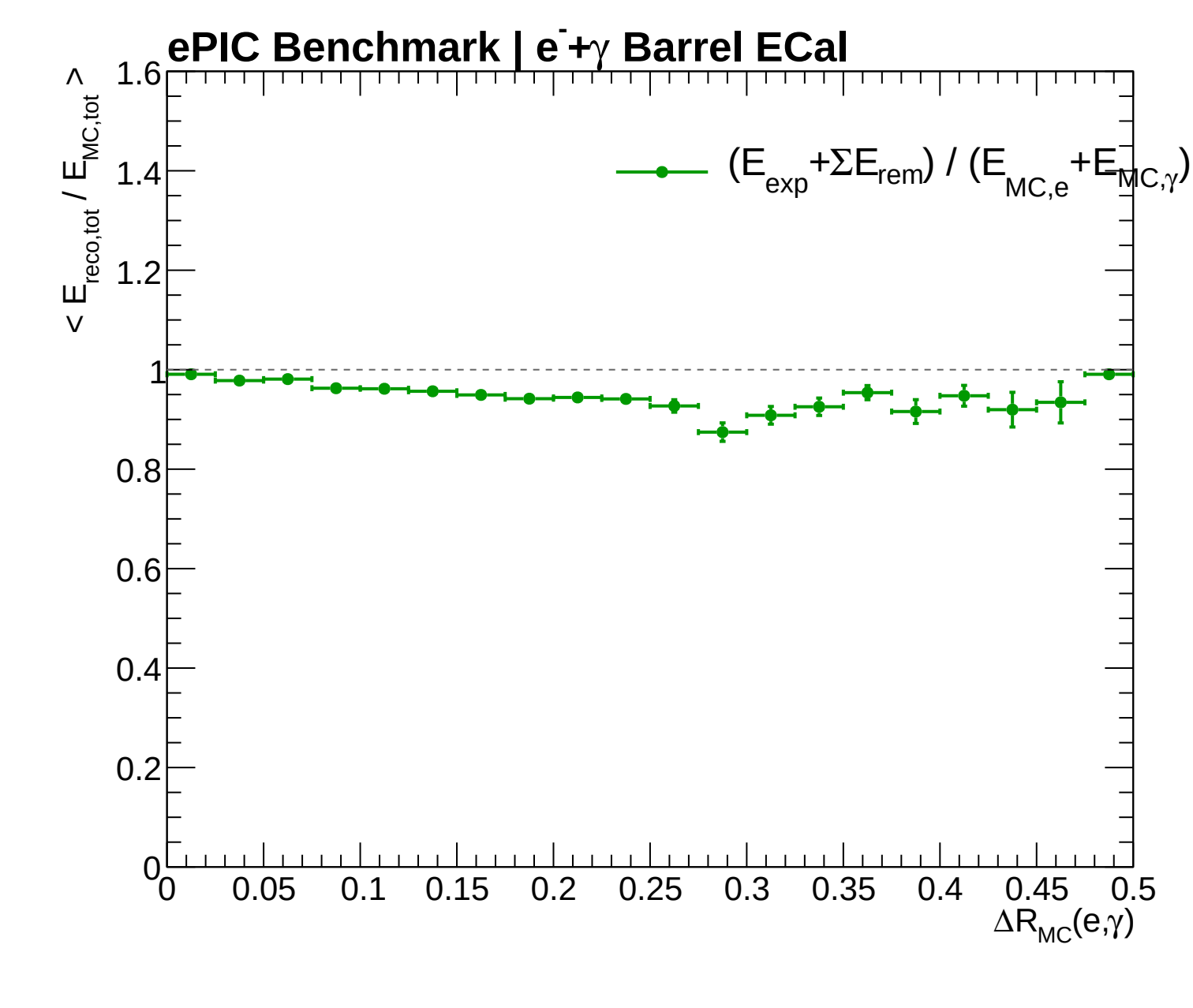

ePIC Benchmark | e⁻+γ Barrel ECal
Δϕ [rad]
0.3
0.2
0.1
0
–0.1
–0.2
–0.3
–0.3
–0.2
–0.1
0
0.1
0.2
0.3
Δη (rem - exp)
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0

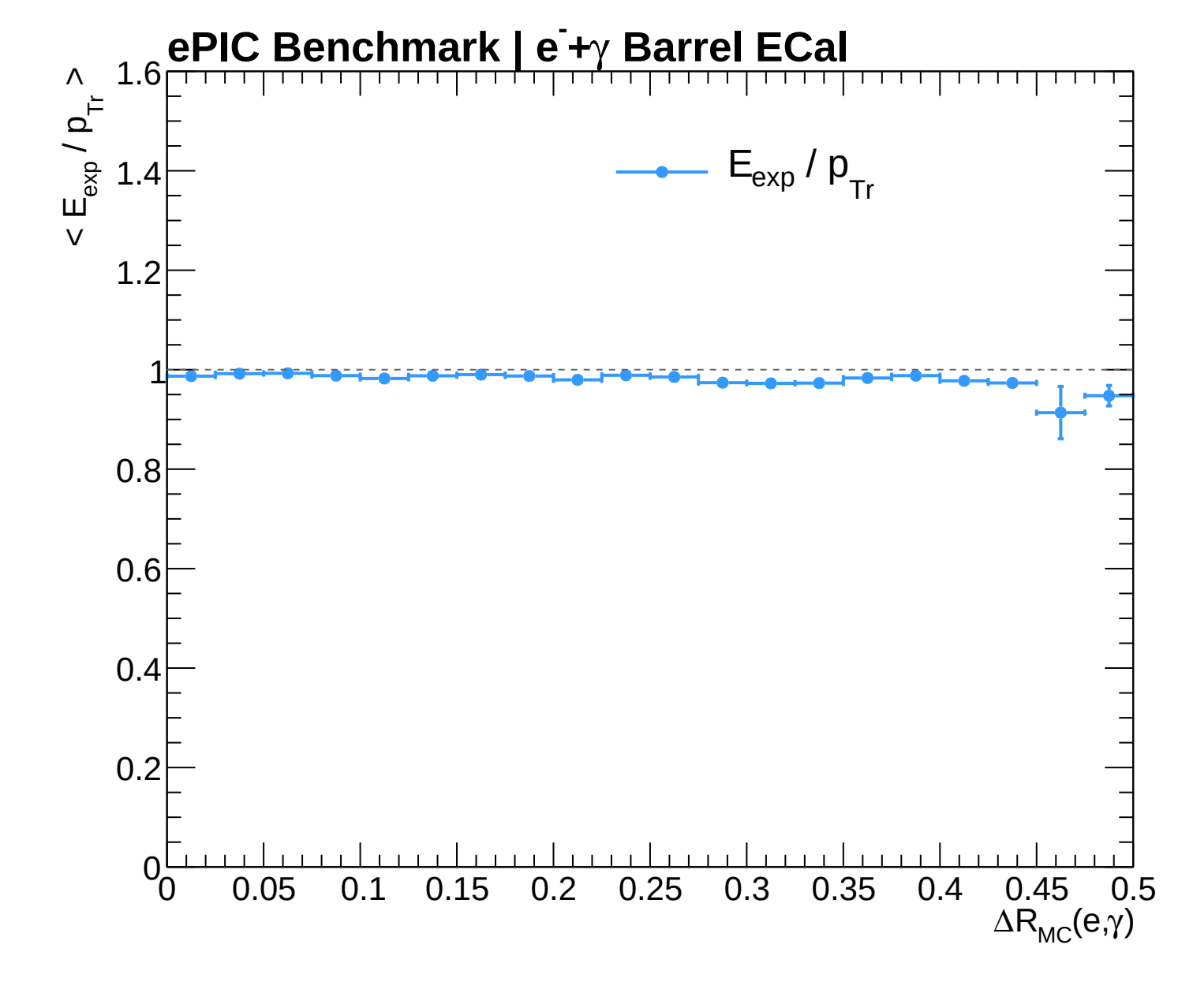

ePIC Benchmark | e⁻+γ Barrel ECal
Events
3000
2500
2000
1500
1000
500
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
R_ηφ(remnant, expected)

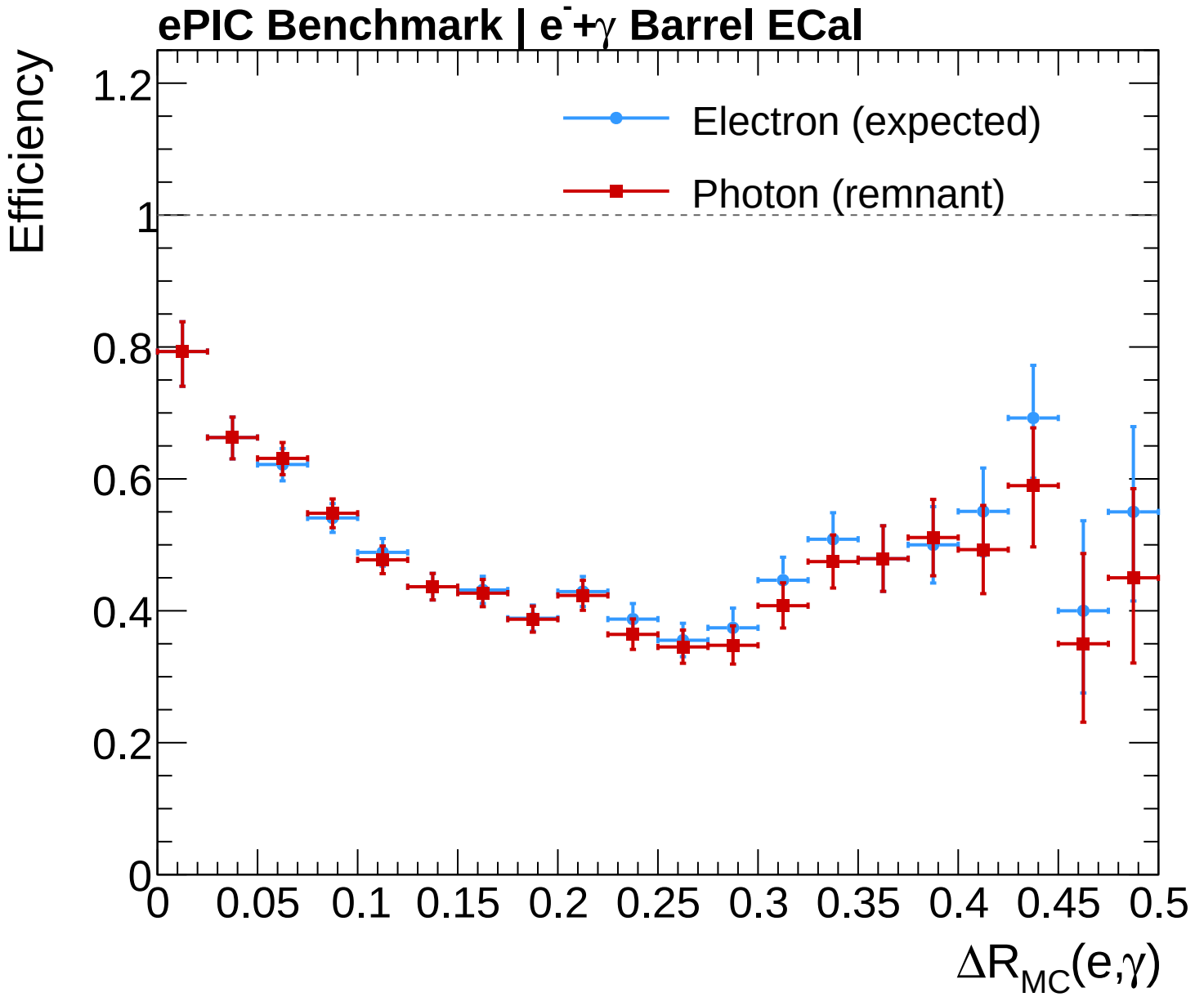

ePIC Benchmark | e⁻+γ Barrel ECal
Black: mean profile
E_exp / E_MC,e
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
20
40
60
80
100
120
140

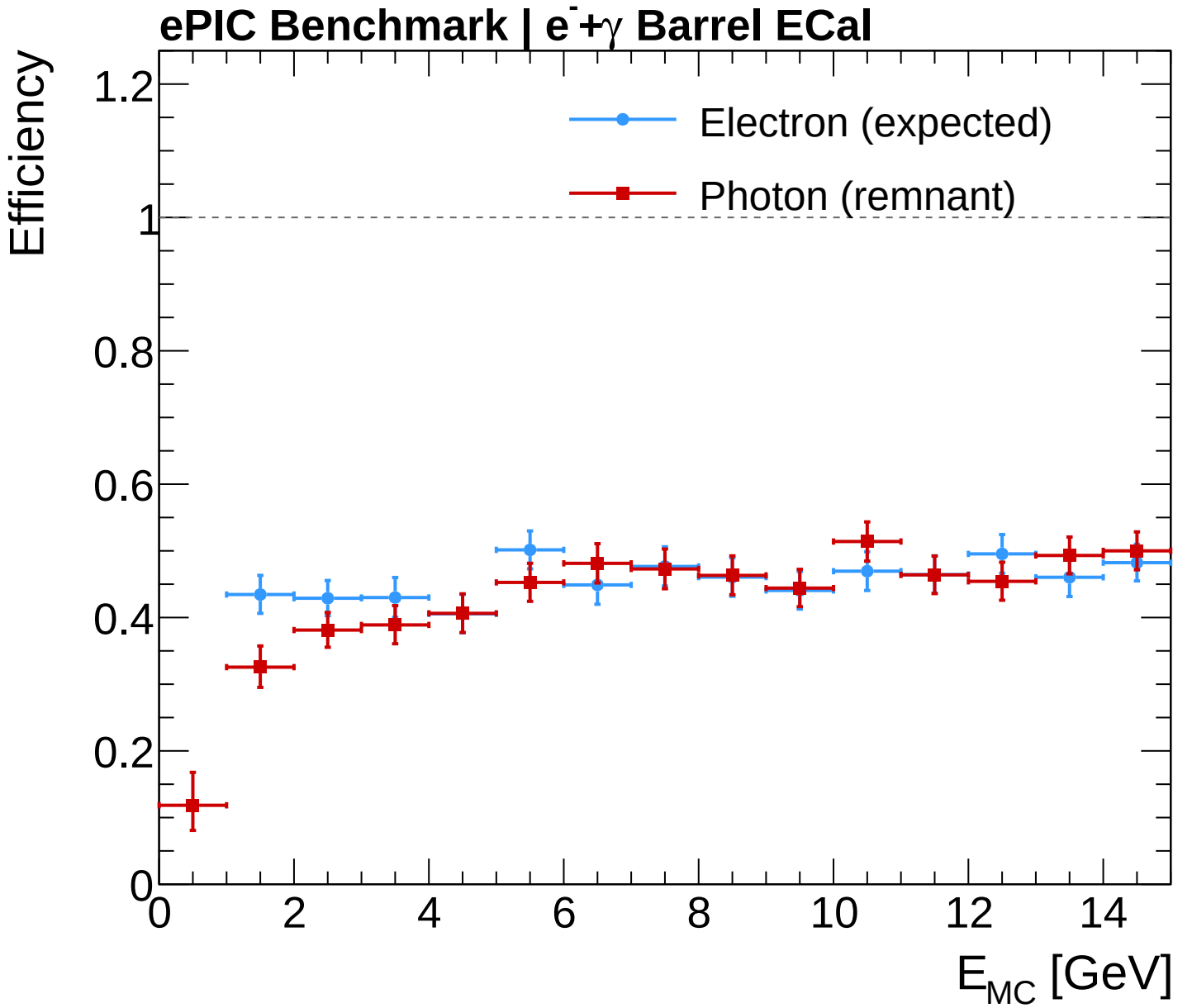

ePIC Benchmark | e⁻+γ Barrel ECal
Black: mean profile
ΣE_rem / E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
10
20
30
40
50

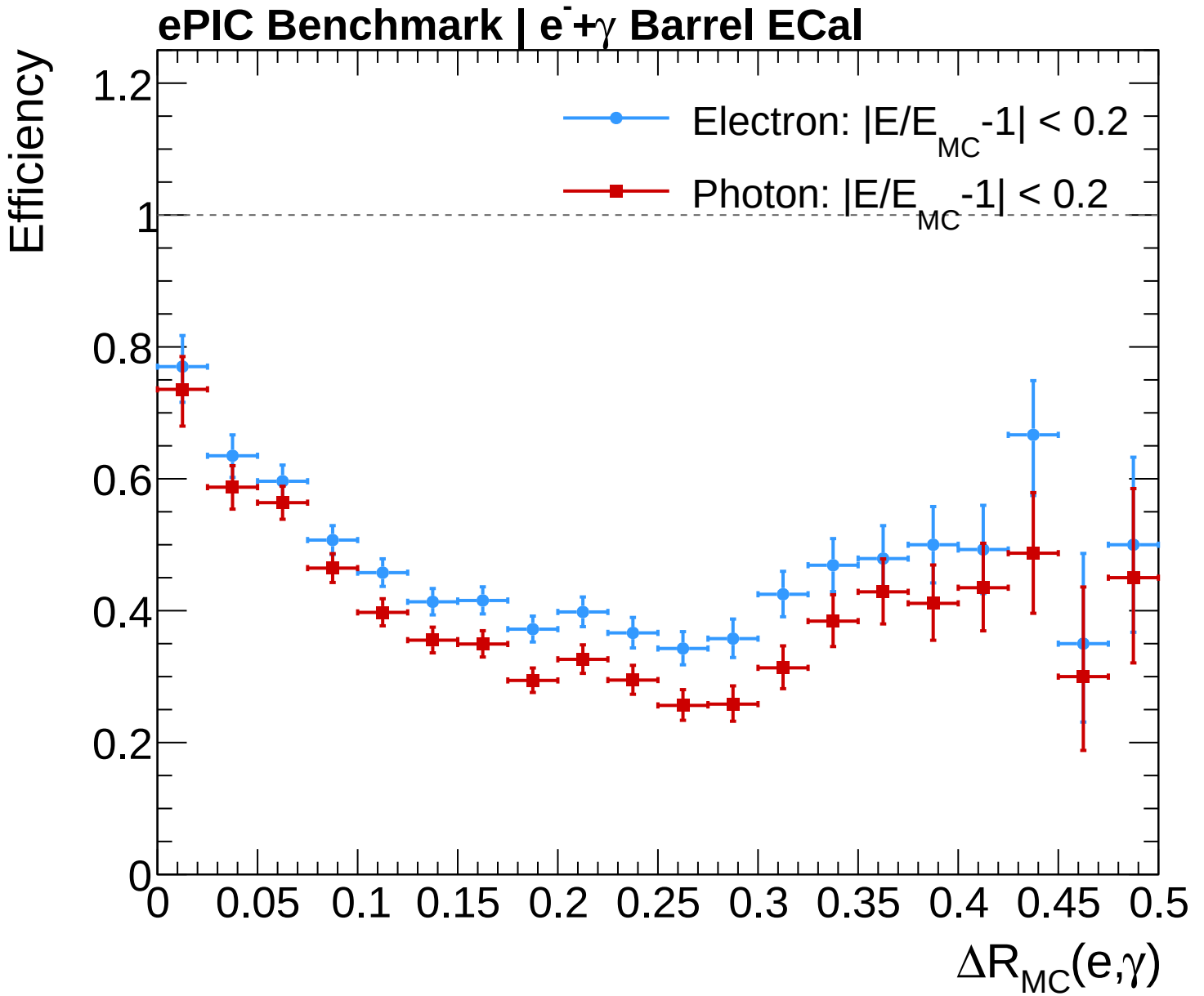

ePIC Benchmark | e⁻+γ Barrel ECal
< E_reco / E_MC >
Electron: E_exp/E_MC,e
Photon: ΣE_rem/E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5


ePIC Benchmark | e⁻+γ Barrel ECal
< E_reco,tot / E_MC,tot >
(E_exp+ΣE_rem) / (E_MC,e+E_MC,γ)
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ Barrel ECal
< E_exp / p_Tr >
E_exp / p_Tr
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ Barrel ECal
Efficiency
Electron (expected)
Photon (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ Barrel ECal
Efficiency
Electron (expected)
Photon (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
E_MC [GeV]


ePIC Benchmark | e⁻+γ Barrel ECal
Efficiency
Electron: |E/E_MC -1| < 0.2
Photon: |E/E_MC -1| < 0.2
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)